

AHMED MOUSSAOUI

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Corvallis, OR (Open to Relocation)

SUMMARY

Mechanical engineering student at Oregon State University (Robotics Control Option) with hands-on experience building vision-guided manipulation systems, autonomous aircraft, and embedded control software, across the full stack from CAD and fabrication to ROS2 and controls. Seeking a robotics, automation, or mechanical design role where I can own projects end to end.

EDUCATION

Oregon State University
B.S. in Mechanical Engineering, Robotics Control Option

Corvallis, OR
Expected June 2026

TECHNICAL SKILLS

CAD and Design: SolidWorks, Fusion 360, Autodesk Inventor, GD&T, DFM, Part and Assembly Design
Fabrication: 3D Printing (FDM), Rapid Prototyping, CNC Machining, Soldering, Wiring and Electrical Assembly, Machine Shop
Robotics: ROS2, MoveIt2, ArduPilot, Mission Planner, Motion Planning, Systems Integration, Embedded Systems
Programming: Python, C++, MATLAB, Arduino, OpenCV, YOLO, Docker, Linux, GitHub
AI and ML: AI API Integration, Computer Vision (OpenCV, YOLO), Pre-trained Model Deployment, Model Fine-tuning
Platforms: Linux, Raspberry Pi, Raspberry Pi OS

WORK EXPERIENCE

Robotics Lab Engineer Feb. 2025 – Present
OSU College of Civil Engineering
Corvallis, OR

- Led a 2–3 person team designing and deploying automation systems for a civil engineering research lab, automating repetitive testing with a 7-DOF robotic manipulator.
- Built a vision-guided pick-and-place pipeline from scratch, cutting manual technician task time by 75% through automated object detection, grasp pose generation, and motion planning.
- Presented system capabilities and progress to prospective investors on behalf of the research group.
- Containerized the system with Docker for reproducible deployment across lab environments.

Undergraduate Learning Assistant, ME 203 Sept. 2025 – Present
OSU MIME Department
Corvallis, OR

- Taught Python for engineering data analysis, numerical methods, and visualization to 25+ students per term.
- Led lab sessions and one-on-one guidance, partnering with faculty to refine course materials.

PROJECTS

Autonomous RC Aircraft | *ArduPilot, Mission Planner, Avionics, Fabrication* Sept. 2024 – June 2025

- Led the autonomous sub-team on a 6–10 person DBF capstone, building the aircraft from scratch including frame, wiring harness, and avionics integration.
- Achieved the first successful autonomous test flights in the team's history, across multiple test days.

SAMI Humanoid Robot | *ROS2, Pygame, Fusion 360, Raspberry Pi* Spring 2025

- Designed and fabricated the full robot head enclosure and screen mount in Fusion 360 on a 3–5 person team, and built an animated face display with real-time ROS2 emotion-state control.
- Delivered a live audience demo of three modes — pose mirroring, dancing, and emotional display — with continuous reliable operation.

Automated Security Door System | *C++, Arduino, Fusion 360, 3D Printing* OSU ME 351

- Designed and 3D printed the full mechanical enclosure, integrating servo actuation, linkages, and sensors to halve part count and improve manufacturability.
- Built a custom metal-detector circuit from scratch and wrote a 5-state C++ control system with jam detection, forced-entry detection, and manual override.

LEADERSHIP

Head Coach, Esports Club Sept. 2024 – Apr. 2025
Oregon State University
Corvallis, OR

- Coached and managed around 20 players across two competitive Valorant teams, improving in-game coordination through structured communication and teamwork drills.

Vice President, LEGO Club Nov. 2024 – Apr. 2025
Oregon State University
Corvallis, OR

- Co-founded the club with faculty support and grew membership to 20+ students in one term through weekly build meetings and hands-on engineering activities.